

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3– Vector Database & Semantic Search Benchmark

Course Code:	CSA65	Course Title:	Generative AI and Large Scale Models
CO Assessed:	CO3 – Precision (BL2, BL3)	Assessment:	Assessment Tool 3 – Vector Database & Semantic Search Benchmark
Weightage:	20% of CIA	Total Marks:	2 * 50 = 100 (1 Question1)
CO3 Statement:	<i>Develop intelligent applications using Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts.(BL3, BL4)</i>		
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Section / Batch:	2024	Date:	

Instructions to Students

- Select/answer the assigned question. Each question carries **50 marks**. Duration 60mins
- Implement the solution using **Python**.
- Use an appropriate embedding model such as Sentence Transformers or Hugging Face.
- Use **FAISS and/or ChromaDB** as specified in the question.
- Demonstrate the complete pipeline:
Document → Preprocessing → Embedding → Vector Storage → Semantic Search → Retrieval → Analysis
- Execute the program and display meaningful outputs.
- Compare FAISS and ChromaDB where required.
- Provide appropriate graphs/tables for benchmark questions.
- Explain the architecture and design decisions.
- Submit working code and demonstrate the complete implementation.

Code of Conduct

I N.Somashekar Naidu(192472347)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.Somashekar Naidu_____

SECTION A – VECTOR EMBEDDINGS & SEMANTIC SEARCH

18.Semantic Search vs Keyword Search

Develop a semantic-search system for a collection of academic documents. Implement both keyword-based search and embedding-based semantic search. Use queries with different wording but similar

1. Assessment Requirement

Develop a persistent ChromaDB semantic-search application. Insert documents and embeddings, terminate the application, restart it, and perform retrieval. Demonstrate that the collection remains available and analyze why persistence is important.

2. Aim

To build a ChromaDB-based semantic search application that persists its vector collection to disk, and to verify that inserted documents remain retrievable after the application process is terminated and restarted.

3. Objectives

- Create a dataset of documents to be indexed for semantic search.
- Generate vector embeddings using the Sentence Transformer model all-MiniLM-L6-v2.
- Create a ChromaDB PersistentClient collection and insert documents with embeddings.
- Terminate the client/process to simulate an application shutdown.
- Restart the application by creating a fresh client pointed at the same persistence directory.
- Verify the collection count is unchanged and perform a semantic query.
- Analyze why persistence matters for real-world RAG/semantic-search systems.

4. Experimental Setup

Total Documents	5,000
Batch Size (insertion)	100
Embedding Model	all-MiniLM-L6-v2
Vector Database	ChromaDB (PersistentClient)
Persistence Directory	./chroma_persistent_store
Collection Name	persistent_docs

5. Methodology

- Generate a synthetic technical-document dataset (5,000 documents across multiple CS topics).
- Convert all documents into vector embeddings using Sentence Transformers.
- Instantiate chromadb.PersistentClient(path=...) so data is written to disk rather than kept only in memory.
- Insert documents, embeddings, and metadata into the collection in batches of 100.
- Delete the client object / end the process, simulating application termination.
- Create a brand-new PersistentClient pointed at the same directory (simulated restart).
- Reconnect to the existing collection with get_collection() and check collection.count().
- Run a semantic query against the reloaded collection and inspect the retrieved results.

6. Architecture / Pipeline

Documents → Embeddings → PersistentClient.add() → Disk-backed store (SQLite + segment files) → [Process terminates] → New PersistentClient() → get_collection() → Verify count → Query → Retrieval

The key architectural difference from an in-memory ChromaDB client (chromadb.Client()) is that PersistentClient writes its index and metadata to the given directory on every insert, so the collection is not lost when the Python process exits. Reconnecting to the same path in a new process re-opens the existing store instead of creating an empty one.

7. Implementation

The complete, runnable implementation is given below. It is organized into two phases mirroring the two runs of the application: Phase 1 creates the collection and inserts 5,000 embedded documents; Phase 2 simulates a fresh application start by opening a new client against the same persistence directory, verifying the count, and performing a semantic query.

```
"""
CO3 - AT3 - Q18: ChromaDB Persistence
Persistent semantic-search application demonstrating that a ChromaDB
collection survives process termination and restart.

Pipeline:
Documents -> Embeddings -> Persistent ChromaDB Collection ->
(simulate shutdown) -> Restart client -> Retrieval -> Verification

Run:
    pip install chromadb sentence-transformers
    python chromadb_persistence.py
"""

import time
import random
import chromadb
from sentence_transformers import SentenceTransformer

PERSIST_DIR = "./chroma_persistent_store"
COLLECTION_NAME = "persistent_docs"
TOTAL_DOCS = 5000
EMBED_MODEL = "all-MiniLM-L6-v2"

TOPICS = [
    "database indexing", "neural networks", "cloud computing", "cyber security",
    "operating systems", "computer networks", "software engineering",
    "data structures", "machine learning", "distributed systems",
    "compiler design", "information retrieval", "natural language processing",
    "computer vision", "big data analytics",
]

def build_dataset(n):
    docs, ids, metadatas = [], [], []
    for i in range(n):
        topic = random.choice(TOPICS)
        docs.append(
            f"Document {i}: An overview of {topic} covering core concepts, "
            f"practical applications, and recent advances relevant to {topic}."
        )
        ids.append(f"doc_{i}")
        metadatas.append({"topic": topic, "doc_index": i})
    return docs, ids, metadatas

def phase_1_create_and_insert():
    """Simulates the first application run: create collection, insert data, exit."""
    print("=" * 60)
    print("PHASE 1 - FIRST RUN: CREATE, EMBED, INSERT, PERSIST")
    print("=" * 60)
```

```

print(f"Building dataset of {TOTAL_DOCS} documents...")
docs, ids, metadatas = build_dataset(TOTAL_DOCS)

print(f"Loading embedding model: {EMBED_MODEL}")
model = SentenceTransformer(EMBED_MODEL)

t0 = time.time()
embeddings = model.encode(docs, show_progress_bar=True, batch_size=64)
embed_time = time.time() - t0
print(f"Embedding generation time: {embed_time:.4f} seconds")

# PersistentClient writes to disk at PERSIST_DIR instead of living in RAM only.
client = chromadb.PersistentClient(path=PERSIST_DIR)
collection = client.get_or_create_collection(name=COLLECTION_NAME)

t0 = time.time()
batch_size = 100
for start in range(0, TOTAL_DOCS, batch_size):
    end = min(start + batch_size, TOTAL_DOCS)
    collection.add(
        ids=ids[start:end],
        documents=docs[start:end],
        embeddings=embeddings[start:end].tolist(),
        metadatas=metadatas[start:end],
    )
insert_time = time.time() - t0
print(f"Batch insertion time: {insert_time:.4f} seconds")
print(f"Collection count immediately after insert: {collection.count()}")

# No explicit close() call is required - PersistentClient flushes to disk
# (SQLite + Parquet segment files) as writes happen. Deleting the client
# object / ending the process is enough to prove persistence on reload.
del client
print("Phase 1 complete. Process/client terminated.\n")
return embed_time, insert_time

def phase_2_restart_and_retrieve(query_text="How does machine learning work?"):
    """Simulates a completely fresh process: reconnect to the same directory."""
    print("=" * 60)
    print("PHASE 2 - SIMULATED RESTART: RECONNECT AND QUERY")
    print("=" * 60)

    # A brand-new PersistentClient pointed at the same path re-opens the
    # existing on-disk store - nothing is re-inserted.
    client = chromadb.PersistentClient(path=PERSIST_DIR)
    collection = client.get_collection(name=COLLECTION_NAME)

    count_after_restart = collection.count()
    print(f"Collection count after restart: {count_after_restart}")
    assert count_after_restart == TOTAL_DOCS, "Persistence check FAILED"
    print("Persistence check PASSED: all documents survived restart.\n")

    model = SentenceTransformer(EMBED_MODEL)
    query_embedding = model.encode([query_text]).tolist()

    t0 = time.time()
    results = collection.query(query_embeddings=query_embedding, n_results=5)
    query_time = time.time() - t0

    print(f"Query: '{query_text}'")
    print(f"Retrieval time: {query_time:.4f} seconds")
    print("Top results:")
    for doc_id, doc, dist in zip(
        results["ids"][0], results["documents"][0], results["distances"][0]
    ):
        print(f" [{doc_id}] (distance={dist:.4f}) {doc[:80]}...")

    return count_after_restart, query_time

```

```

if __name__ == "__main__":
    embed_time, insert_time = phase_1_create_and_insert()

    print(">>> Simulating application termination and restart <<<\n")

    count_after_restart, query_time = phase_2_restart_and_retrieve()

    print("=" * 60)
    print("PERSISTENCE BENCHMARK SUMMARY")
    print("=" * 60)
    print(f"Total documents inserted : {TOTAL_DOCS}")
    print(f"Embedding generation time : {embed_time:.4f} seconds")
    print(f"Batch insertion time : {insert_time:.4f} seconds")
    print(f"Collection count on reload : {count_after_restart}")
    print(f"Query retrieval time : {query_time:.4f} seconds")
    print("Conclusion: ChromaDB PersistentClient retains all data across restarts.")

```

8. How to Run and Capture Evidence

- Install dependencies: `pip install chromadb sentence-transformers`
- Run: `python chromadb_persistence.py`
- Phase 1 output shows embedding generation time and batch insertion time, and the collection count immediately after insert.
- The script then prints '>>> Simulating application termination and restart <<<' and Phase 2 begins with a fresh PersistentClient.
- Phase 2 output shows the collection count after restart (should equal 5,000) and the result of a sample semantic query.
- For a stronger persistence demonstration, Phase 1 and Phase 2 can also be run as two separate script invocations / two separate terminal sessions against the same `./chroma_persistent_store` directory.
- Take a screenshot of the full console output and paste it into Section 9 below, then transfer the measured values into the results table in Section 9.

9. Actual Execution Results

[Run the script above and paste your real console output / screenshot here before submission. Insert the measured values in the table below.]

Measurement	Actual Result
Total Documents	[fill in]
Embedding Generation Time	[fill in]
Batch Insertion Time (Phase 1)	[fill in]
Collection Count Immediately After Insert	[fill in]
Collection Count After Simulated Restart	[fill in]
Query Retrieval Time (Phase 2)	[fill in]
Persistence Check Result	[PASSED / FAILED]

10. Database Verification

Persistence is verified by asserting that `collection.count()` returns the same value (5,000) both immediately after Phase 1's insertion and again after Phase 2 reconnects to the store in a completely new client instance. Because Phase 2 never calls `add()` again, an unchanged count is direct evidence that the data was read back from disk rather than re-created.

11. Analysis and Interpretation

Once the results above are filled in, the analysis should confirm two things: first, that the collection count reported after the simulated restart matches the count reported right after insertion, demonstrating that no data was lost when the process terminated; second, that the semantic query executed against the reloaded collection returns relevant results, demonstrating that the embeddings — not just the raw text — survived the restart intact and remain usable for similarity search.

12. Why Persistence Is Important

- Re-embedding a large document collection on every application start is computationally expensive; persistence avoids repeating that cost.
- Production RAG systems must survive server restarts, deployments, and crashes without losing their knowledge base.
- Persistence decouples the lifetime of the vector index from the lifetime of any single process, enabling multiple processes to share one store.
- It allows incremental updates: new documents can be added over time without rebuilding the entire index.
- It is a prerequisite for backing up, versioning, or migrating a vector database.

13. Conclusion

A persistent ChromaDB semantic-search application was implemented using `chromadb.PersistentClient`. Documents and their embeddings were inserted in Phase 1, and the application was terminated. In Phase 2, a fresh client was created against the same persistence directory, and the collection was shown to still contain all inserted documents and to support semantic queries. This confirms that ChromaDB's persistence mechanism reliably retains both data and embeddings across restarts, which is essential for any production-grade retrieval system.

14. Demo / Viva Explanation

“I built a ChromaDB semantic search app using a `PersistentClient` pointed at a local directory instead of an in-memory client. I generated embeddings for my documents with the `all-MiniLM-L6-v2` model and inserted them in batches. After confirming the collection count, I terminated the client to simulate the application shutting down. I then created a brand-new client pointed at the same directory to simulate a fresh application start, reconnected to the collection, and confirmed the document count was unchanged. I also ran a semantic query against the reloaded collection to show the embeddings were still usable. This demonstrates that ChromaDB persists both the raw data and the vector index to disk, which is critical for production RAG systems that can't afford to rebuild their index on every restart.”

15. Submission Checklist

- Documents and embeddings created.
- `PersistentClient` collection created and populated.
- Application/client termination simulated.
- Fresh client reconnected to the same persistence directory.
- Collection count verified unchanged after restart.
- Semantic query executed successfully after restart.
- Actual execution console output / screenshot captured and inserted in Section 9.
- Analysis, conclusion, and demo explanation included.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Database Implementation	30%	Correctly builds and queries both FAISS and ChromaDB with accurate understanding of indexing	Builds both with minor issues	Only one database is built and queried correctly	Neither database functions correctly
Retrieval Relevance & Ranking	25%	Retrieval results are relevant and correctly ranked for all test queries	Mostly relevant results across queries	Inconsistent relevance across queries	Results are largely irrelevant
Comparative Analysis	20%	Thoughtful comparison of speed/accuracy trade-offs between the two databases	Reasonable comparison of the two databases	Superficial comparison with little insight	No meaningful comparison attempted
Benchmark Presentation	15%	Clear benchmark results presented (e.g. table or chart)	Results presented adequately	Results presented unclearly	No clear results presented
Methodology Documentation	10%	Clearly documents methodology and test queries used	Adequate documentation of methodology	Minimal documentation	No documentation of methodology

Marks Evaluation:

Question No.	Pipeline Functionality(30%)	Answer Relevance & Groundedness (25%)	Query Robustness(20%)	End-to-End Demo(15%)	Architecture Explanation (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						